Chapter 5. Potential Problems in Regression

Lecture 6-2 (assumptions, non-Gaussianity, outliers and robust regression, breakdown point, measurement error)

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1 Assumptions and Common Issues

The linear model relies on: linearity; normally distributed errors; independent errors; constant variance; no outliers or high-leverage points; small collinearity.

Outliers are observations with unusually large residuals; **high-leverage points** have unusual predictor values; **correlated errors** violate independence.

Kind of problem	Examples
data problems	outliers, measurement error, missing data
model problems	omitted-variable bias, misspecification, multicollinearity
error problems	non-normality, heteroscedasticity, autocorrelation

Remark: diagnostics in brief

Nonlinearity: residuals versus fitted values show a pattern (fix: transformations, polynomials, splines). **Heteroscedasticity**: a funnel in the residual plot (fix: transform Y, e.g. $\log Y$ or $\sqrt{Y}$, or weighted least squares). **Correlated errors**: tracking in residuals over time; standard errors become too small. **Leverage**: $h_{ii} = x_i^\top (X^\top X)^{-1} x_i$, average $(p+1)/n$; flag $h_{ii} \gg (p+1)/n$. **Collinearity**: variance inflation factor $\mathrm{VIF}(\hat{\beta}_j) = 1/(1 - R^2_{X_j|X_{-j}})$; values above 5–10 are problematic.

2 Non-Gaussianity: Normality Is Not an Issue

Motivation. Normality of errors is often listed as a regression assumption, but the estimates are unbiased without it and the CLT makes inference approximately valid for large samples. What matters more is independence and constant variance.

Theorem: exact distribution under Gaussian errors

If $y = X\beta + \varepsilon$ with $\varepsilon \sim N(0, \sigma^2 I)$ and X of full column rank, then

$$\hat{\beta} \sim N(\beta, \sigma^2 (X^\top X)^{-1}), \qquad \hat{\beta}_j \sim N(\beta_j, \sigma^2 v_j),$$

where v_j is the jth diagonal element of $(X^\top X)^{-1}$. Replacing σ by $\hat{\sigma} = \sqrt{\mathrm{RSS}/(n-p)}$ (p columns of X including the intercept),

$$t_j = \frac{\hat{\beta}_j - \beta_j}{\hat{\sigma}\sqrt{v_j}} \sim t_{n-p},$$

which gives tests and confidence intervals.

Proof [Supplement]. $\hat{\beta} = \beta + (X^\top X)^{-1} X^\top \varepsilon$ is a linear map of a Gaussian vector, with mean β and covariance $\sigma^2 (X^\top X)^{-1}$. The residual vector $(I - H)\varepsilon$ is independent of $\hat{\beta}$ (uncorrelated jointly Gaussian), and $\mathrm{RSS}/\sigma^2 = \varepsilon^\top (I - H)\varepsilon / \sigma^2 \sim \chi^2_{n-p}$ since $I - H$ is a projection of rank $n - p$. The ratio of a standard normal and an independent $\sqrt{\chi^2_{n-p}/(n-p)}$ is t_{n-p}. ■

2.1 Large-sample inference

If the errors are not normal, $\hat{\beta}$ is not exactly normal, but by the CLT

$$\hat{\beta} \;\dot{\sim}\; N(\beta, \sigma^2 (X^\top X)^{-1}) \quad \text{for large } n,$$

(provided errors are i.i.d. with finite variance and the design is not dominated by a few points, i.e. $\max_i h_{ii} \to 0$). Hence t-tests and confidence intervals remain approximately valid. Alternatively use **bootstrap inference**: resample the data with replacement, recompute $\hat{\beta}$ for each bootstrap sample, and use the empirical distribution of $\hat{\beta}^*$ for standard errors and intervals (Ch. 3). Resampling pairs (x_i, y_i) is also valid under heteroscedasticity.

3 Outliers and Robust Regression

3.1 Why OLS is sensitive

OLS minimizes $\sum_{i=1}^{n}(y_i - x_i^\top \beta)^2$. Squaring residuals amplifies large errors, so a single extreme observation can strongly influence the fitted model. This motivates robust regression.

3.2 Alternative objectives

- **Least absolute deviation (LAD)**: $\hat{\beta} = \arg\min_\beta \sum_i |y_i - x_i^\top \beta|$; the conditional median (Ch. 2 analogue: absolute loss is minimized by the median).
- **Huber regression**: $\hat{\beta} = \arg\min_\beta \sum_i \rho_c(r_i)$, $r_i = y_i - x_i^\top \beta$,

$$\rho_c(r) = \begin{cases} \frac{1}{2}r^2, & |r| \le c, \\ c|r| - \frac{1}{2}c^2, & |r| > c. \end{cases}$$

- **Least trimmed squares (LTS)**: $\hat{\beta} = \arg\min_\beta \sum_{i=1}^{h} r_{(i)}^2$, using only the h smallest squared residuals ($r_{(1)}^2 \le r_{(2)}^2 \le \cdots$).

Derivation: Huber as iteratively reweighted least squares

ρ_c is convex with derivative $\psi_c(r) = r$ for $|r| \le c$ and $c\,\mathrm{sign}(r)$ otherwise. The estimating equation $\sum_i \psi_c(r_i)x_i = 0$ equals the weighted normal equation $\sum_i w_i r_i x_i = 0$ with $w_i = \min(1, c/|r_i|)$, so one iterates $\beta \leftarrow (X^\top W X)^{-1} X^\top W y$. Large residuals get weights $c/|r_i| < 1$: their influence is bounded by c, whereas under OLS it is unbounded. Usually $c = 1.345\hat{s}$ for a robust scale $\hat{s}$ (95% efficiency under normal errors).

3.3 Breakdown point

Definition: breakdown point

The (finite-sample) breakdown point of an estimator is the smallest fraction of contaminated observations that can make the estimate arbitrarily large (break it down).

OLS	LAD	Huber M-estimator	least median of squares	least trimmed squares
0	0	0	0.5	≈ 0.5

OLS can be destroyed by a single outlier (breakdown $1/n \to 0$): moving one y_i to infinity moves $\hat{\beta}$ to infinity. LAD and Huber resist outliers in y, but a single bad *leverage point* (outlying x_i) can still break them, hence breakdown 0. High-breakdown estimators (LMS: minimize the median squared residual; LTS with $h \approx n/2$) tolerate nearly 50% contamination — more than half bad points can never be distinguished from good ones, so 0.5 is the maximum.

4 Measurement Error

Suppose the true model is $Y = \beta_0 + \beta_1 X + \varepsilon$, but the predictor is measured with error:

$$\tilde{X} = X + u, \qquad u \text{ independent of } (X, \varepsilon),\ \mathbb{E}u = 0.$$

Regressing Y on the observed $\tilde{X}$ gives a biased estimator.

Theorem: attenuation bias

$$\hat{\beta}_1 \xrightarrow{p} \beta_1 \frac{\mathrm{Var}(X)}{\mathrm{Var}(X) + \mathrm{Var}(u)} = \lambda \beta_1, \qquad 0 < \lambda \leq 1.$$

Proof [Supplement]. The OLS slope converges to $\mathrm{Cov}(\tilde{X}, Y)/\mathrm{Var}(\tilde{X})$. Here $\mathrm{Cov}(X + u, \beta_0 + \beta_1 X + \varepsilon) = \beta_1 \mathrm{Var} X$ and $\mathrm{Var}(\tilde{X}) = \mathrm{Var} X + \mathrm{Var} u$. ■

The slope is shrunk toward zero (**attenuation bias**); λ is the reliability ratio. Measurement error in Y only, in contrast, inflates the error variance but does not bias the slope. In multiple regression, error in one predictor also biases the coefficients of correlated predictors.

4.1 Remedies

Method	Idea
instrumental variables	use a variable Z correlated with X but independent of the measurement error: $\hat{\beta}_1^{\mathrm{IV}} = \widehat{\mathrm{Cov}}(Z, Y)/\widehat{\mathrm{Cov}}(Z, \tilde{X})$
repeated measurements	average m noisy measurements, reducing $\mathrm{Var} u$ to $\mathrm{Var} u/m$; replicates also estimate $\mathrm{Var} u$ so that $\hat{\beta}_1/\hat{\lambda}$ corrects the bias
errors-in-variables models	model the measurement process explicitly (e.g. Deming regression)
SIMEX	add extra artificial noise of increasing size, observe how the estimate changes, and extrapolate back to zero measurement error
validation data	use a subset where the true X is observed to calibrate $\tilde{X}$ (regression calibration)

Summary

- Regression problems arise from the data (outliers, measurement error, missing data), the model (omitted variables, misspecification, collinearity) and the error structure (heteroscedasticity, autocorrelation).
- Normality is often not critical: the CLT gives large-sample inference and the bootstrap is an alternative.
- Outliers motivate robust regression (LAD, Huber, LTS); the breakdown point measures robustness, and only high-breakdown methods resist leverage points.
- Error in a predictor causes attenuation bias toward zero; IV, replicates, SIMEX or validation data correct it.

5 Selected exercises

Motivation. The diagnostics box and the robustness discussion state several facts without proof: the range and average of the leverages, the VIF formula, the effect of one outlier on OLS, and the consistency of the instrumental-variable estimator. The exercises prove them.

Exercise 5.1 (Properties of leverage)

Let $X \in \mathbb{R}^{n\times q}$ have full column rank q and first column $\mathbf{1}$, and $H = X(X^\top X)^{-1}X^\top$ with diagonal entries h_{ii}.
(a) Show that $0 \le h_{ii} \le 1$ and $\sum_i h_{ii} = q$, so the average leverage is q/n ($q = p+1$).
(b) Show that $h_{ii} \ge 1/n$.
(c) Show that $h_{ii} = 1$ forces $\hat{y}_i = y_i$ for every response vector y.

Solution [Supplement].

(a) H is symmetric and idempotent, so $h_{ii} = (H^2)_{ii} = \sum_j h_{ij}^2 = h_{ii}^2 + \sum_{j\neq i} h_{ij}^2 \ge h_{ii}^2$, which gives $0 \le h_{ii} \le 1$. Also $\sum_i h_{ii} = \mathrm{tr}H = \mathrm{tr}((X^\top X)^{-1}X^\top X) = q$.

(b) $J = \mathbf{1}\mathbf{1}^\top/n$ projects onto $\mathrm{span}(\mathbf{1}) \subseteq \mathrm{col}(X)$, so $HJ = JH = J$ and $H - J$ is symmetric and idempotent. The argument of (a) applied to $H - J$ gives $(H-J)_{ii} \ge 0$, i.e. $h_{ii} \ge 1/n$.

(c) If $h_{ii} = 1$, the identity in (a) gives $\sum_{j\neq i} h_{ij}^2 = 0$, so $h_{ij} = 0$ for $j \neq i$ and $\hat{y}_i = \sum_j h_{ij}y_j = y_i$: the fit is forced through point i, whatever its response. ■

Exercise 5.2 (The variance inflation factor)

In the model of Exercise 5.1 with $\mathrm{Var}\varepsilon = \sigma^2 I$, let column $j \ge 2$ be x and the remaining columns W (including $\mathbf{1}$). Let R_j^2 be the R^2 of regressing x on W and $S_{jj} = \sum_i (x_i - \bar{x})^2$. Show that

$$\mathrm{Var}(\hat{\beta}_j) = \frac{\sigma^2}{(1-R_j^2)\,S_{jj}} = \mathrm{VIF}(\hat{\beta}_j)\cdot\frac{\sigma^2}{S_{jj}}.$$

Solution [Supplement].

Step 1 (a Schur complement). Order the columns as $[W, x]$, so

$$X^\top X = \begin{pmatrix} W^\top W & W^\top x \\ x^\top W & x^\top x \end{pmatrix}, \qquad s = x^\top x - x^\top W(W^\top W)^{-1}W^\top x = x^\top(I - P_W)x,$$

with P_W the projection onto $\mathrm{col}(W)$. Full column rank gives $x \notin \mathrm{col}(W)$, so $s = \|(I-P_W)x\|^2 > 0$. The vector $v = s^{-1}(-(W^\top W)^{-1}W^\top x;\ 1)$ satisfies $X^\top X v = s^{-1}(0;\ x^\top x - x^\top W(W^\top W)^{-1}W^\top x) = (0;\ 1)$, so v is the last column of $(X^\top X)^{-1}$ and its last entry is $1/s$.

Step 2. s is the residual sum of squares of regressing x on W. Since W contains $\mathbf{1}$, the total sum of squares of that regression is S_{jj}, so $s = (1-R_j^2)S_{jj}$.

Step 3. $\mathrm{Var}(\hat{\beta}) = \sigma^2(X^\top X)^{-1}$; its entry for column j is σ^2/s. If x is uncorrelated with the other predictors ($R_j^2 = 0$) this is σ^2/S_{jj}, so $1/(1-R_j^2)$ is exactly the inflation caused by collinearity. ■

Exercise 5.3 (One outlier breaks OLS)

In the setting of Exercise 5.1, replace y_i by $y_i + t$. Show that the new estimate is $\hat{\beta}(t) = \hat{\beta} + t(X^\top X)^{-1}x_i$, that $\|\hat{\beta}(t)\| \to \infty$ as $|t| \to \infty$, and that the fitted value at x_i moves by $t\,h_{ii}$ with $h_{ii} \ge 1/n$.

Solution [Supplement]. $\hat{\beta}$ is linear in y, so $\hat{\beta}(t) = (X^\top X)^{-1}X^\top(y + te_i) = \hat{\beta} + t(X^\top X)^{-1}x_i$. The vector $u = (X^\top X)^{-1}x_i$ is nonzero, since $(X^\top X)^{-1}$ is invertible and $x_i \neq 0$ (its first entry is 1). Hence $\|\hat{\beta}(t)\| \ge |t|\,\|u\| - \|\hat{\beta}\| \to \infty$. The fitted value at x_i changes by $x_i^\top tu = t\,h_{ii}$, and $h_{ii} \ge 1/n > 0$ by Exercise 5.1. One contaminated observation out of n suffices, so the breakdown point of OLS is $1/n$. (Compare the median, which needs about half the sample: Exercise 1.3.) ■

Exercise 5.4 (Error in Y, and instrumental variables)

Let $(X_i, Y_i, \dots)$ be i.i.d. with finite second moments, $Y = \beta_0 + \beta_1 X + \varepsilon$, $\mathrm{Var}X > 0$, $\mathbb{E}\varepsilon = 0$, $\mathrm{Cov}(X, \varepsilon) = 0$.
(a) If only $\tilde{Y} = Y + v$ is observed, with $\mathbb{E}v = 0$ and $\mathrm{Cov}(X, v) = 0$, show that the OLS slope of $\tilde{Y}$ on X is consistent for β_1.
(b) If only $\tilde{X} = X + u$ is observed (as in the attenuation theorem) and Z satisfies $\mathrm{Cov}(Z, X) \neq 0$, $\mathrm{Cov}(Z, u) = \mathrm{Cov}(Z, \varepsilon) = 0$, show that $\hat{\beta}_1^{\mathrm{IV}} = \widehat{\mathrm{Cov}}(Z, Y)/\widehat{\mathrm{Cov}}(Z, \tilde{X}) \xrightarrow{p} \beta_1$.

Solution [Supplement]. By the law of large numbers, sample means of the variables and of their pairwise products converge to their expectations, so every sample covariance converges in probability to the population covariance; a ratio converges to the ratio of the limits when the denominator limit is nonzero (continuous mapping).

(a) The slope is $\widehat{\mathrm{Cov}}(X, \tilde{Y})/\widehat{\mathrm{Var}}(X) \xrightarrow{p} \mathrm{Cov}(X, \beta_0 + \beta_1 X + \varepsilon + v)/\mathrm{Var}X = \beta_1$. The error of the observed regression is $\varepsilon + v$, so only the noise level (and hence the standard errors) increases.

(b) $\mathrm{Cov}(Z, Y) = \beta_1 \mathrm{Cov}(Z, X) + \mathrm{Cov}(Z, \varepsilon) = \beta_1 \mathrm{Cov}(Z, X)$ and $\mathrm{Cov}(Z, \tilde{X}) = \mathrm{Cov}(Z, X) + \mathrm{Cov}(Z, u) = \mathrm{Cov}(Z, X) \neq 0$. The ratio of the limits is β_1. The measurement error disappears because Z is uncorrelated with it, while OLS divides by $\mathrm{Var}\tilde{X} = \mathrm{Var}X + \mathrm{Var}u$. ■